

EDUCATION

Northeastern University

Boston, USA

Ph.D. in Marine and Environmental Sciences

2024 - Present

- Advisor: Prof. Cristina Schultz
- Dissertation: Linking Terrestrial Inputs, Coastal Benthic Fluxes, and Carbon Dynamics: A Multi-Scale Study from the Plum Island Estuary to the Gulf of Maine

China University of Geosciences (Beijing)

Beijing, China

M.Eng. in Geological Engineering (U.S. Equivalent: M.Sc.)

2018 - 2021

Zhejiang University

Hangzhou, China

B.Eng. in Environmental Engineering

2013 - 2017

POSITIONS

Chinese Academy of Sciences

Nanjing, China

Research Associate, Nanjing Institute of Geography and Limnology

2021 - 2024

China University of Geosciences (Beijing)

Beijing, China

Lab Manager, MC-ICP-MS Laboratory

2019 - 2020

PUBLICATIONS

1. Wu, Y., Chen, S., **Ye, Q.**, Jiang, L., Chang, Y., Huang, W., Liu, C., Ma, S., Jiang, R., Fei, D., Xu, H., Shi, X., & Chen, K. (2025). Influence of Coverage and Diversity of Submerged Macrophytes on the Sediment Bacterial Community in Dongshan Bay, Lake Taihu. *Freshwater Biology*, 70(11).
2. Wu, Y., **Ye, Q.**, Chen, S., Chen, K., Ma, S., Chang, Y., & Shi, X. (2024). Response of phytoplankton functional group to spring drought in a large subtropical reservoir. *Journal of Oceanology and Limnology*, 42(5), 1586–1596.
3. Wu, Y., **Ye, Q.**, Chen, S., Jiang, L., Chen, K., Ma, S., Chang, Y., Liu, J., & Shi, X. (2024). Climate changes drive phytoplankton community through complementarity and selection effects in a large mesotrophic reservoir. *Journal of Oceanology and Limnology*, 43(1), 119–133.
4. Chen, S., Jiang, L., Ma, S., Wu, Y., **Ye, Q.**, Chang, Y., Ye, Y. & Chen, K. (2024). Response of a submerged macrophyte (*Vallisneria spiralis*) to water depth gradients and sediment nutrient concentrations. *Science of The Total Environment*, 912, p.169154.
5. Yu, C., Wang, H., Blaustein, R.A., Guo, L., **Ye, Q.**, Fu, Y., Fan, J., Su, X., Hartmann, E.M. & Shen, C. (2022). Pangenomic and functional investigations for dormancy and biodegradation features of an organic pollutant-degrading bacterium *Rhodococcus biphenylivorans* TG9. *Science of The Total Environment*, 809, p.151141.

RESEARCH

Northeastern University

2024 - 2029

Department of Marine and Environmental Sciences

As a Ph.D. student working with Prof. Cristina Schultz, I'm dedicated to evaluating the influence of benthic processes and terrestrial input on coastal nutrient cycles and water quality. In particular, I am:

- Compiling and integrating data from long-term observational records, hydrological datasets, and high-resolution ocean biogeochemistry models to build a comprehensive database for analyzing nutrient fluxes, oxygen dynamics, and sediment properties.
- Investigating the relationships among key environmental factors in PIE, particularly the role of benthic processes in regulating nutrient cycling and oxygen availability.
- Comparing patterns across estuaries by examining how PIE's nutrient fluxes, oxygen dynamics, and sediment properties differ from other northeastern U.S. estuaries, while exploring connectivity between these coastal systems.
- Assessing the impact of extreme events (e.g., storms, heatwaves) on estuarine dynamics and developing guidelines for future long-term monitoring and data integration to improve our understanding of estuarine resilience.

Chinese Academy of Sciences

2021 - 2024

Nanjing Institute of Geography and Limnology

As a research associate working with Prof. Kaining Chen and Prof. Xiaoli Shi, I focused on exploring the relationships between phytoplankton and the water column in lakes and reservoirs. In these projects, I:

- Investigated the mechanisms behind algal blooms caused by changes in nutrient concentrations, meteorological and hydrological conditions with long-term data series.
- Studied the impact of global warming on drinking water sources.

China University of Geosciences (Beijing)

2018 - 2021

Institute of Earth Sciences

The title of my master's thesis is An Improved Method for Precise Selenium Isotope Determination. Under the guidance of Prof. Jianming Zhu, I:

- Improved the testing method for precise selenium isotope determination.
- Tested and reported the selenium isotope ratios of a series of standard samples.

Zhejiang University

2016 - 2017

College of Environmental and Resource Sciences

The title of my undergrad's thesis is Hormesis Response of Polychlorinated Biphenyls to *Escherichia Coli*. Under the guidance of Prof. Chaofeng Shen, I:

- Established a microplate based, high throughput test method for *E. coli* growth.
- Identified the types of PCBs that can show the hormesis response on *E. coli* growth and investigated the mechanisms.

INTERESTS

- Relationships between climate change, human activities, and aquatic system.
- Environmental complex systems modeling using **R** and **Python**.
- Large-scale dataset modeling and simulation.
- Environmental sustainability and environmental geochemistry.

WORKSHOPS	Ocean Carbon & Biogeochemistry <i>Invited BECS Workshop Speaker</i> - Delivered a 45-min hands-on tutorial on running the 1-D CBED (Carbon, Benthic Ecosystems, and Diagenesis) sediment biogeochemistry model in R. - Guided participants to select global ocean sites and generate key diagnostics, including sediment nutrient depth profiles, top boundary exchange conditions, and bottom nutrient flux time-series plots. - Provided plotting tutorial slides, covered model workflow, parameterization basics, and visualization best practices to support integration among oceanographic modelers.	Princeton, USA 2025
TEACHING	Northeastern University <i>Teaching Assistant, Department of Marine and Environmental Sciences</i> - Served as the instructor of 2 sections of Lab for Biostatistics (ENVR 2501), an undergraduate course, each with 19 students. - Presented an one-hour lecture per section on advanced statistical methods in R (Bayesian / frequentist approaches, generalized linear models, mixed-effects models, multivariate statistics, and ensemble modeling), followed by a two-hour Q&A session every week. - Provided hands-on guidance, working closely with students on real-world data analysis to ensure they developed strong practical skills in R.	Spring 2025
MENTORING	Chinese Academy of Sciences <i>Mentor, Nanjing Institute of Geography and Limnology</i> - Served as a research mentor for two undergraduate students from Imperial College London during a research internship. - Trained students in the collection of water, phytoplankton, and zooplankton samples from field sites. Supervised laboratory analysis of nutrient concentrations using standard chemical methods. Guided students through preliminary environmental data processing and interpretation.	2022
	China University of Geosciences (Beijing) <i>Mentor, MC-ICP-MS Laboratory</i> - Served as a senior graduate mentor for junior master's students in the isotope geochemistry laboratory. - Provided training on the complete workflow of sample preparation, including dissolution and purification of geological and environmental samples. Conducted one-on-one tutorials on the use of MC-ICP-MS for precise isotopic analysis.	2019 – 2020
ACADEMIC SERVICES	Freshwater Biology <i>Peer Reviewer</i> Evaluated and provided feedback on manuscripts related to freshwater ecosystems, assessing scientific rigor, methodology, and ecological significance.	2024 - 2025
SCHOLARSHIPS	- Full Scholarship Ph.D. Program, Northeastern University. - Second-Class Academic Scholarship, China University of Geosciences.	2024 2019, 2020

ACTIVITIES

Northeastern University

2024 - 2025

Member of the Ph.D. Student Council, College of Science

- Serve as a liaison between the Dean's Office and Ph.D. students, ensuring their concerns and suggestions were represented.
- Build community among Ph.D. students across different departments through events and programs, fostering collaboration.
- Provide a platform for sharing ideas, addressing issues, and encouraging constructive dialogue within the College.

SKILLS

Programming & Software

- Proficient in **R**, **Python**, and **MATLAB** for data analysis, modeling and other computational tasks.
- Experienced in large-scale modeling in high-performance computing environments.
- Skilled with ArcGIS, MIKE Zero, Origin, L^AT_EX, Microsoft Office, Adobe Creative Suite.

Laboratory & Field Work

- Proficient in field navigation for locating sampling sites using portable GPS devices.
- Proficient in sample collection and preservation (water, sediment, phytoplankton, zooplankton, aquatic plants, etc.).
- Proficient in chemical laboratory skills, including sample preparation and testing.
- Proficient in ultra-clean laboratory procedures such as isotope sample preparation.
- Skilled in the operation and maintenance of experimental instrumentation including UV-Spectrophotometer, Atomic Fluorescence Spectrometer (AFS), Inductively Coupled Plasma Mass Spectrometer (ICP-MS), and Multi-Collector ICP-MS (MC-ICP-MS).

Languages

- Chinese: Native.
- English: Full professional proficiency.